ORF307: Optimization

Precept: #1

Agenda

- Setting up your Python environment (Codespaces and Docker)
- Linear systems of equations
- Inner and Outer products of vectors
- Matrix multiplication
- Trace of a matrix
- Norms
- Cauchy-Schwartz inequality

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW: ....
- Other announcements on blackboard.

Linear systems

Linear algebra provides a way to compactly represent linear system

Example: Consider the linear system of two variables and two equations

$$\begin{aligned} 4x_1 - 5x_2 &= -13 \\ -2x_1 + 3x_2 &= 9 \end{aligned}$$

It can be compactly written in matrix-vector form as follows:

$$Ax = b$$

where

$$A = \begin{bmatrix} 4 & -5 \\ -2 & 3 \end{bmatrix} \in \mathbb{R}^{2\times 2}, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2, \quad b = \begin{pmatrix} -13 \\ 9 \end{pmatrix} \in \mathbb{R}^2$$

Inner products

The inner product of two vectors $c, x \in \mathbb{R}^n$ is a scalar and is defined as

$$c^T x = \sum_{i=1}^{n} c_i x_i \quad \in \mathbb{R}$$

Example:

$$c = \begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}, \quad x = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \qquad c^T x = (0 \;\; 5 \;\; 2) \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} = \sum_{i=1}^{3} c_i x_i = 0 - 5 + 6 = 1$$

Outer products

The outer product of two vectors $c, x \in \mathbb{R}^n$ is an $n \times n$ matrix and is defined as

$$cx^T = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} (x_1 \;\; x_2 \;\; \dots \;\; x_n) = \begin{bmatrix} c_1x_1 & c_1x_2 & \dots & c_1x_n \\ c_2x_1 & c_2x_2 & \dots & c_2x_n \\ \vdots & \vdots & \ddots & \vdots \\ c_nx_1 & c_nx_2 & \dots & c_nx_n \end{bmatrix}$$

Remark: $rank(ab^T) = 1$ for every $a, b \in \mathbb{R}^n$.

This is easy to prove by noting that the columns of the matrix ab^T are scalar multiples of each other.

Remark: The outer product is used in the fundamental definition of the covariance matrix for a random vector.

To convince yourselves consider the 2-dimensional random vector and its mean vector

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \qquad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}.$$

Then the outer product $(X - \mu)(X - \mu)^T$ is the matrix:

$$(X - \mu)(X - \mu)^T = \begin{bmatrix} (X_1 - \mu_1)^2 & (X_1 - \mu_1)(X_2 - \mu_2) \\ (X_2 - \mu_2)(X_1 - \mu_1) & (X_2 - \mu_2)^2 \end{bmatrix}$$

Now, if we take the expected value of each element in the above matrix, we obtain the covariance matrix:

$$\Sigma = \mathbb{E}[(X - \mu)(X - \mu)^T] = \begin{bmatrix} \mathbb{E}[(X_1 - \mu_1)^2] & \mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)] \\ \mathbb{E}[(X_2 - \mu_2)(X_1 - \mu_1)] & \mathbb{E}[(X_2 - \mu_2)^2] \end{bmatrix}$$

The diagonal elements are the variances of the random variables X_1 and X_2 and the off-diagonal elements are their covariances (more details about this topic are covered in "ORF309-Probability and Stochastic Systems")

Matrix multiplication

If you have two matrices $A \in \mathbf{R}^{m \times n}$ and $B \in \mathbf{R}^{n \times p}$ then their product $C = AB$ is an $m \times p$ matrix ($C \in \mathbb{R}^{m \times p}$) whose (i, j) element is defined as

$$C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

The matrix multiplication can be viewed in at least three different ways

1. Inner product view:

$$C = AB = \begin{bmatrix} — & \boldsymbol{a_1}^T & — \\ — & \boldsymbol{a_2}^T & — \\ & \vdots & \\ — & \boldsymbol{a_m}^T & — \end{bmatrix} \begin{bmatrix} | & | & & | \\ \boldsymbol{b^1} & \boldsymbol{b^2} & \dots & \boldsymbol{b^p} \\ | & | & & | \end{bmatrix} = \begin{bmatrix} \boldsymbol{a_1}^T \boldsymbol{b^1} & \boldsymbol{a_1}^T \boldsymbol{b^2} & \dots & \boldsymbol{a_1}^T \boldsymbol{b^p} \\ \boldsymbol{a_2}^T \boldsymbol{b^1} & \boldsymbol{a_2}^T \boldsymbol{b^2} & \dots & \boldsymbol{a_2}^T \boldsymbol{b^p} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{a_m}^T \boldsymbol{b^1} & \boldsymbol{a_m}^T \boldsymbol{b^2} & \dots & \boldsymbol{a_m}^T \boldsymbol{b^p} \end{bmatrix}$$

2. Outer product view:

$$C = AB = \begin{bmatrix} | & | & & | \\ \boldsymbol{a^1} & \boldsymbol{a^2} & \dots & \boldsymbol{a^n} \\ | & | & & | \end{bmatrix} \begin{bmatrix} — & \boldsymbol{b_1}^T & — \\ — & \boldsymbol{b_2}^T & — \\ & \vdots & \\ — & \boldsymbol{b_n}^T & — \end{bmatrix} = \sum_{i=1}^{n} \boldsymbol{a^i} \boldsymbol{b_i}^T$$

3. Matrix-vector products view:

$$C = AB = A \begin{bmatrix} | & | & & | \\ \boldsymbol{b^1} & \boldsymbol{b^2} & \dots & \boldsymbol{b^p} \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ A\boldsymbol{b^1} & A\boldsymbol{b^2} & \dots & A\boldsymbol{b^p} \\ | & | & & | \end{bmatrix}$$

Trace of a matrix

The trace of a matrix $A \in \mathbb{R}^{n \times n}$ is the sum of its diagonal elements

$$Tr(A) = \sum_{i=1}^{n} A_{ii}$$

Exercise: Show that $Tr(AB) = Tr(BA)$ for all matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$.
Proof: Note that $AB \in \mathbf{R}^{m \times m}$ and $BA \in \mathbb{R}^{n \times n}$.

$$Tr(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m} (\sum_{j=1}^{n} A_{ij} B_{ji}) = \sum_{j=1}^{n} (\sum_{i=1}^{m} B_{ji} A_{ij}) = \sum_{j=1}^{n} (AB)_{jj} = Tr(BA)$$

Property connecting the outer product and the trace of two n-dimensional vectors:

You can easily show that

$$Tr(ab^T) = a^T b$$

where $a, b \in \mathbb{R}^n$.

Remark: The trace of a covariance matrix, Σ, is the sum of the variances of all the random variables. This sum, known as the **total variance**, gives a single value that represents the overall spread of the data.

Norms

Informally, norms measure the length of a vector $x \in \mathbb{R}^n$. Formally, a norm of a vector $x \in \mathbb{R}^n$ is any function $f : \mathbb{R}^n \to \mathbb{R}$ that satisfies the following **four** properties

1. **Non-negativity**: $f(x) \geq 0, \forall x \in \mathbb{R}^n$
2. **Definitelness**: $f(x) = 0 \iff x = 0$
3. **Homogeneity**: $f(tx) = |t|f(x),\ \ \forall t \in \mathbb{R}, x \in \mathbb{R}^n$
4. **Triangle inequality**: $f(x+y) \leq f(x) + f(y),\ \ \forall x, y \in \mathbb{R}^n$

Examples of norms: The following are the most commonly used norms

- $L2$ or Euclidean norm:
$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$$
- $L1$ norm:
$$\|x\|_1 = \sum_{i=1}^{n} |x_i|$$
- L_∞ norm:
$$\|x\|_\infty = \max_{i=1,\dots,n} \{|x_i|\}$$

Cauchy-Schwartz inequality

It connects the inner product of two vectors with their Euclidean norms

$$x^T y \leq \|x\|_2 \|y\|_2$$